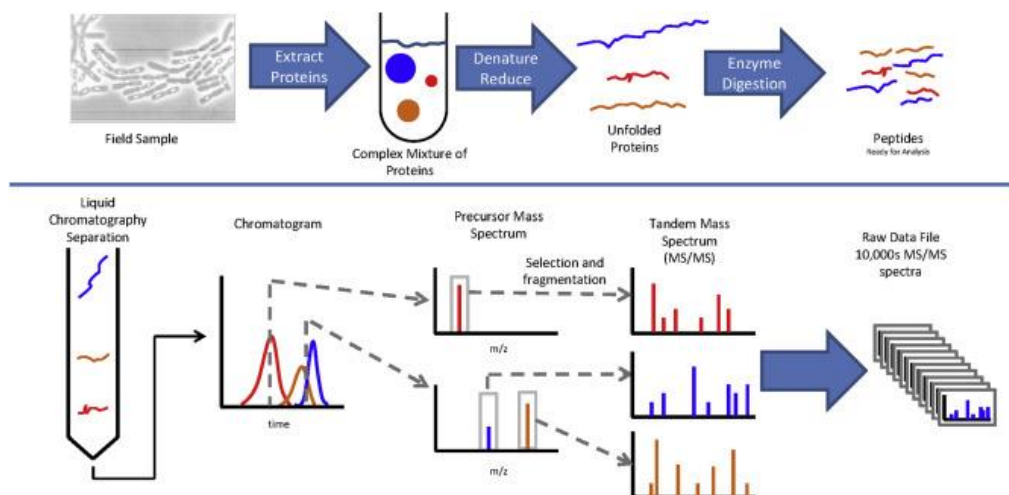


Title: Liquid Chromatography-Mass Spectrometry for bacteria recognition

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Samples contains certain bacteria for the sick patients, and other bacteria for healthy patients. With Liquid Chromatography-Mass Spectrometry process, the samples are decomposed to extract peptides. This separation takes time so different peptides are extracted little by little through time.



During this separation process, mass spectrums are captured through time (Retention Time axis on the figure below). These spectrums are represented by histograms of the intensity of each peptide. Actually the histograms are not built at the precision of each peptide, they are a bit less precise by aggregating the peptides depending on their mass index (m/z axis on the figure below), so one peak of an histogram may correspond to several different peptides.

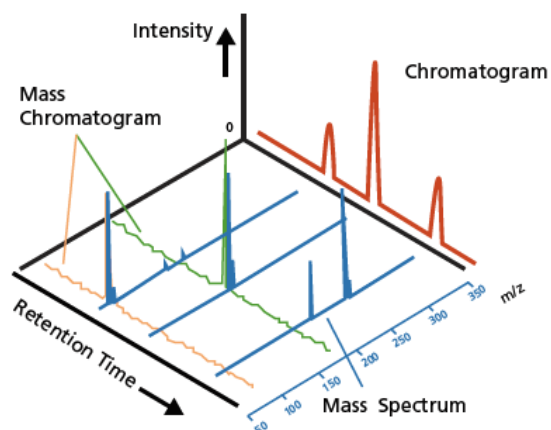


Figure: A typical LC chromatogram (red) and a LC-MS mass chromatogram and mass spectrum (blue).

In this project, samples come from human urine to detect bladder infections of pregnant women.

In the figure below, there is a schema of a first baseline we have made but the results are not good enough.

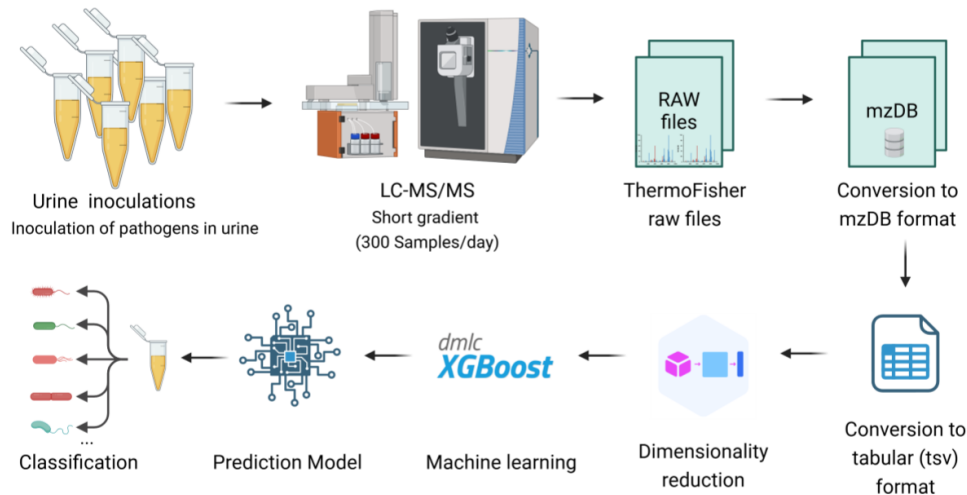


Figure 1: Pipeline for rapid bacterial identification using LC-MS/MS and machine learning. Urine samples inoculated with bacterial species undergo ultra-fast gradient LC-MS/MS analysis, generating raw data that is converted into mzDB format for optimized storage. The files are then processed into TSV format using a binning strategy to extract key mass spectrometry features (m/z and retention time). A structured matrix is generated for machine learning, where an XGBoost model has been trained to classify bacterial species based on extracted features. A confidence threshold strategy ensures high reliability, and the final output provides a classification table indicating the detected bacterial species.

Your objective in the project is to outperform our baseline using deep networks.

There are three problems to solve:

1. Each sample is a huge histogram (of size 100,000 !), so you will have to find a solution to reduce the dimensions of the samples.
2. The acquisitions have not been made in just one same day for all the patients. So we know there is a “Batch Effect” in these data, meaning that, samples from sick patients are different from samples from other sick patients decomposed through LC-MS another day, which are also different from the sample from again other sick patients acquired another day. This is the same for samples from healthy patients. Each acquisition is a bit different from the previous ones.
3. Classification of 28 bacteria types.

This project will be done in collaboration with the team of Professor Arnaud Droit in CHU University Laval, Quebec, Canada.

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